

Date: 30/07/2025

Task 2: Implement conditional, control and looping statement

a. Temperature Alert System (if-elif)

AIM: TO Write a python program that classifies temperature as "Too Cold", "Comfortable", or "Too Hot" using conditional (if-elif-else) statements.

Algorithm:-

1. start the program
2. Accept temperature input from the user
3. If temperature < 18 , print "Too Cold"
4. Else if temperature is between 18 and 25 (inclusive) print "Comfortable"
5. Else, print "Too Hot"
6. End

program:

```
# Temperature Alert System
```

```
temperature = float(input("Enter the room temperature:"))
```

```
if temperature < 18:
```

```
    print("Too Cold")
```

```
elif 18 <= temperature <= 25:
```

```
    print("Comfortable")
```

```
else:
```

```
    print("Too Hot")
```

OUTPUT (Example):

Enter the room temperature: 20
comfortable

Enter the room temperature: 28

Too hot

Enter the room temperature: 16

Too cold.



NAME	DATE	TIME	MARKS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

VEL TECH - CSE	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

RESULT:

The program correctly identifies the room temperature range and prints an appropriate alert.

b. Password Retry system (While loop)

AIM:

TO implement a password retry system using a while loop that allows a maximum of 3 attempts.

Algorithm.

1. Start.
2. Set correct password as "admin123"
3. Set attempt counter to 0
4. While attempts < 3 :
 - ASK the user for password input
 - If correct, print "Access Granted" and exit loop
5. If attempts == 3, Print "Access Denied"
6. End

Program:

```
# Password Retry system
correct_password = "admin123"
attempts = 0
while attempts < 3:
    entered_password = input("Enter Password:")
    if entered_password == correct_password:
        print("Access Granted")
        break
    else:
        print("Incorrect Password. Try Again.")
        attempts += 1
if attempts == 3:
```


OUTPUT (example)

Enter password : test

Incorrect password - Try Again.

Enter password : admin

Incorrect password - Try Again

Enter password : admin 123

Access Granted.

VRL TECH - CSE	
EX NO.	
PERFORMANCE (S)	
RESULT AND ANALYSIS (S)	
VIVA VOCE (S)	
RECORD (S)	

C. Factorial Finder (For loop)

AIM:

TO Write a python program that calculates the factorial of a number using a for loop.

Algorithm:-

1. start
2. Input a number from the user
3. Initialize result as 1
4. Loop from 1 to number (inclusive), multiply result by each value
5. print the result
6. End.

Program:-

```
# Factorial Finder
```

```
number = int(input("Enter a number: "))
```

```
factorial = 1
```

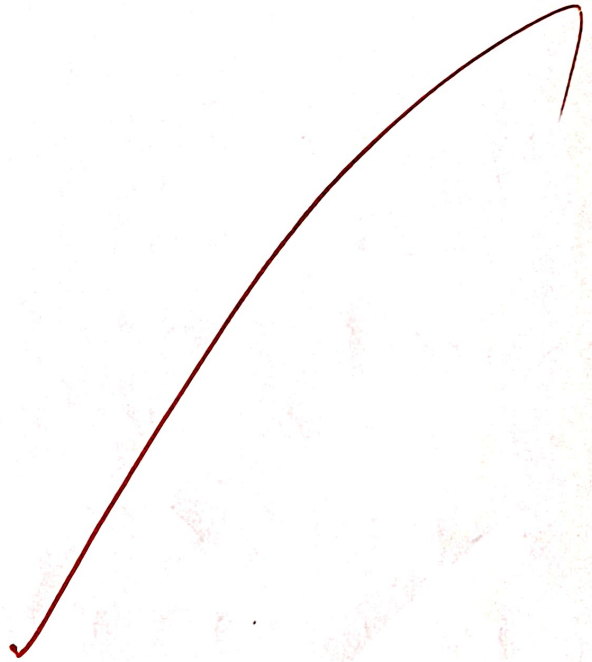
```
for i in range(1, number + 1):
```

```
    factorial *= i
```

```
print("Factorial of", number, "is", factorial)
```


OUTPUT (Example):

Enter a number $x=5$
factorial of 5 is 120.



VRL TECH - GST	
EX. NO.	
PERFORMANCE (S)	
RESULT AND ANALYSIS (S)	

VEL TECH - CSE	
EX NO.	2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	15
SIGNATURE DATE	13/8

RESULT:

The program correctly calculates the factorial of the entered number using a for loop.